SACHIIN M

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Education

B.E. Computer Science & Engineering (AIML) CGPA: 7.5 / 10 | 2023–2027
KPR Institute of Engineering & Technology, Coimbatore
Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Operating Systems, DBMS, OOP

Projects

Tessera — Multi-Agent Software Agency with Independent Verification Oct 2026 – Present
Python · FastAPI · SQLite · Nebius Token Factory · NVIDIA Nemotron

- Built for the **Nebius x NVIDIA Global AI Hackathon**; designed a **multi-agent pipeline** (Main, Tech Lead, Developer, Tester) turning a one-line idea into a verified FastAPI app, **11/11 requirements passing** on a demo invoicing app.
- Tester agent writes **black-box tests from the API contract alone**; automated **fix loop** caught and resolved a real validation bug. Sandboxed runner, per-role model routing, token budgeting; open-sourced (MIT) on GitHub.

Adaptive RAG — Final Year Project *(Team Lead, team of 4)* 2026 – Present
Python · RAG · LLMs · Vector Search

- Leading a team of **4** to build an **adaptive RAG** system that adjusts retrieval strategy to the query and grounds LLM answers in source documents, directing architecture and evaluation design against a baseline RAG.

Text-to-SQL Fine-Tuning with LoRA / QLoRA 2026
Python · LoRA · QLoRA · Qwen2.5-Coder · Google Colab (T4 GPU)

- Fine-tuned **Qwen2.5-Coder-1.5B** (LoRA) and **-7B 4-bit** (QLoRA) on a single Colab T4 to convert a question + schema into SQL; raised exact-match on **300 held-out questions** from **49.7%** to **81.3%** (r=16).
- Ran rank/learning-rate ablations (**r=4–64**, 4.6M–73.9M params) — rank had negligible effect, showing fine-tuning mainly taught the dataset's SQL conventions, not new capability.

Work Experience

Project Intern — Payoda, Coimbatore Feb – Jul 2026
PostgreSQL · FastAPI · Node.js · Playwright · HTML · CSS · Agile/Scrum · LLMs

- Designed a relational **PostgreSQL schema of 21 tables** and built core backend services on it, including a shipment API, payment sandbox integration, and wishlist modules.
- Built **Node.js + Playwright** end-to-end test automation for internal apps (AI TestGen, SnapTest, ScriptWeaver), resolving integration issues across tools.
- Delivered a **multi-agent AI system** optimizing LLM token usage and context length, working within an **Agile/Scrum** team through sprints and stand-ups.

Research Intern — Providence University, Taiwan Oct – Nov 2025
CNN · PyTorch · Grad-CAM · Medical Imaging

- Trained a DenseNet-based CNN pneumonia classifier on **7,750 chest X-ray images**, achieving **92.44% accuracy** and **0.94 F1-score**.
- Integrated Grad-CAM heatmaps to localize predictions and generated automated PDF medical reports per patient; resolved a 4:1 class imbalance with SMOTE and image augmentation.

Software Developer Intern — Zettaone Technologies, Bengaluru Jul – Aug 2025
Flask · React Native · OpenCV · EasyOCR · YOLOv8

- Built a PCB component deviation detection system using **YOLOv8** and OpenCV, flagging manufacturing defects in real time via a Flask CV pipeline.
- Built **VISYRE** (React Native + Flask) for QR/barcode decoding and **OCR** text extraction using **EasyOCR**, eliminating manual data entry.

Research Publication & Presentation

Sachiin Megarajan, Khire Rushikesh Ulhas, Ramesh Kumar T, Sutapa Debbarma, "Pneumonia Detection from Chest X-Ray Images using Dense Net," *All India Seminar on AI, Design Thinking and Innovation*, Institution of Engineers (India), Tripura State Centre Jan 2026

Skills

Languages & Backend	Python, SQL, FastAPI, Flask, Node.js, PostgreSQL, REST APIs
ML / DL / CV	PyTorch, CNNs, Grad-CAM, LoRA fine-tuning, model evaluation, OpenCV, YOLOv8, EasyOCR
GenAI	LLM agents, RAG, MCP, multi-agent systems
Tools & Process	Git, Playwright, Google Colab, Agile, Scrum